

YUCHEN DU

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SUMMARY

Ph.D. candidate at Purdue University specializing in AI-driven optimization and transportation systems. Applied Scientist intern at Amazon, where I built graph-enhanced multi-agent LLM systems for e-commerce appeal adjudication. Published in Transportation Research Part C & E, with experience in stochastic optimization, reinforcement learning, simulation.

SKILLS

- **Mathematical Optimization:** Mixed Integer Programming, Heuristics, Stochastic modeling.
- **AI/ML:** Reinforcement Learning, Generative Adversarial Networks, Transformers.
- **Project Experience:** LLM agent systems, last-mile delivery optimization, transportation system simulation.
- **Programming:** Python (Gurobi, PyTorch, LangGraph), C# (Unity, Godot), SQL

EDUCATION

Purdue University, Ph.D. in Industrial Engineering Technology	2023-Expected Dec 2026
University of Michigan, MSc in Data Science track at Industrial and Operations Engineering	2021-2023
South China University of Technology, BSc in Industrial Engineering	2017-2021

EXPERIENCE

Amazon – Applied Scientist Intern Seattle, WA | Aug 2025-Nov 2025

- Deployed a graph-enhanced multi-agent LLM pipeline for seller-appeal adjudication, addressing the need to audit enforcement decisions across scattered operational logs, policy actions, and historical correction patterns; transformed cases into Evidence-Action-Factor-Decision (EAFD) graphs and integrated GraphRAG, retrieval-augmented inference, task decomposition, ReAct planning, and self-reflection to verify evidence, detect evidence gaps, and produce auditable decision support.

PAPERS

Reviewing the Reviewer: Graph-Enhanced LLMs for E-commerce Appeal Adjudication

[arXiv:2603.19267](https://arxiv.org/abs/2603.19267) | First author | Research output from Amazon Applied Scientist internship

- Proposed graph-enhanced LLM agents for evidence-grounded seller-appeal review under the Evidence-Action-Factor-Decision (EAFD) schema; the agents retrieve cross-case correction patterns, apply top-down deductive reasoning to new appeals, and flag evidence conflicts or gaps, achieving 95.8% offline and 96.3% production alignment with human experts versus a 70.8% LLM-only baseline.

EVC-S: A High-Fidelity Simulation Framework for Three-Sided Electric Mobility Markets

Under review at Simulation Modelling Practice and Theory, 2026 | 2nd author

- Co-developed a NYC-scale three-sided eMaaS simulator coupling traveler demand (TLC/HV/FHV trips), mobility-provider EV dispatch/charging/rebalancing, and AFDC/Revel charging-provider constraints (capacity, power, pricing/access hours); benchmarked hybrid backup-provider scenarios via wait-time, station-revenue, and depot-profit metrics.

“A two-stage stochastic optimization model for sidewalk robot food delivery systems”

Published, Transportation Research Part E, 2025 | First author

- Designed a Decision-Making under Uncertainty system for autonomous logistics networks. Implemented a Scalable Meta-heuristic Search (ALNS) algorithm to optimize Resource Allocation across distributed agents, achieving superior system throughput under high-dimensional constraints.

Analytical model for large-scale design of sidewalk delivery robot systems with bounded makespans

Published, Transportation Research Part C, 2025 | 2nd author

- Developed a Continuous Approximation model for fleet sizing and cost optimization in large-scale sidewalk delivery robot (SDR) networks. Validated on 10K+ synthetic NYC data, identified significant cost savings for strategic SDR deployment.

Understanding Social Perception, Interactions, and Safety Aspects of Sidewalk Delivery Robots Using Sentiment Analysis

Published, Transportation Research Record, 2025 | First author

- Applied BERT, LSTM, and topic modeling to analyze 10K+ YouTube comments, identifying key concerns for public perception of SDRs. Provided data-driven policy insights, shaping recommendations for SDR regulations.